CSA 0613-DESIGN AND ANALYSIS OF ALGORITHMS

Comparing The Best, Average, And Worst Case Analysis Of The Quick Sort Algorithm Using Asymptotic Notation

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INTRODUCTION

- Sorting is the process of arranging data in a specific order, such as ascending or descending, to improve data organization and searching efficiency.
- Quick Sort is one of the most efficient and widely used sorting algorithms based on the Divide and Conquer strategy.
- The performance of Quick Sort depends significantly on the choice of the pivot element
- The three important cases considered are:

  Best Case – Balanced partitioning of elements.

  Average Case – Randomly distributed elements resulting in reasonably balanced partitions.

  Worst Case – Highly unbalanced partitions caused by poor pivot selection.

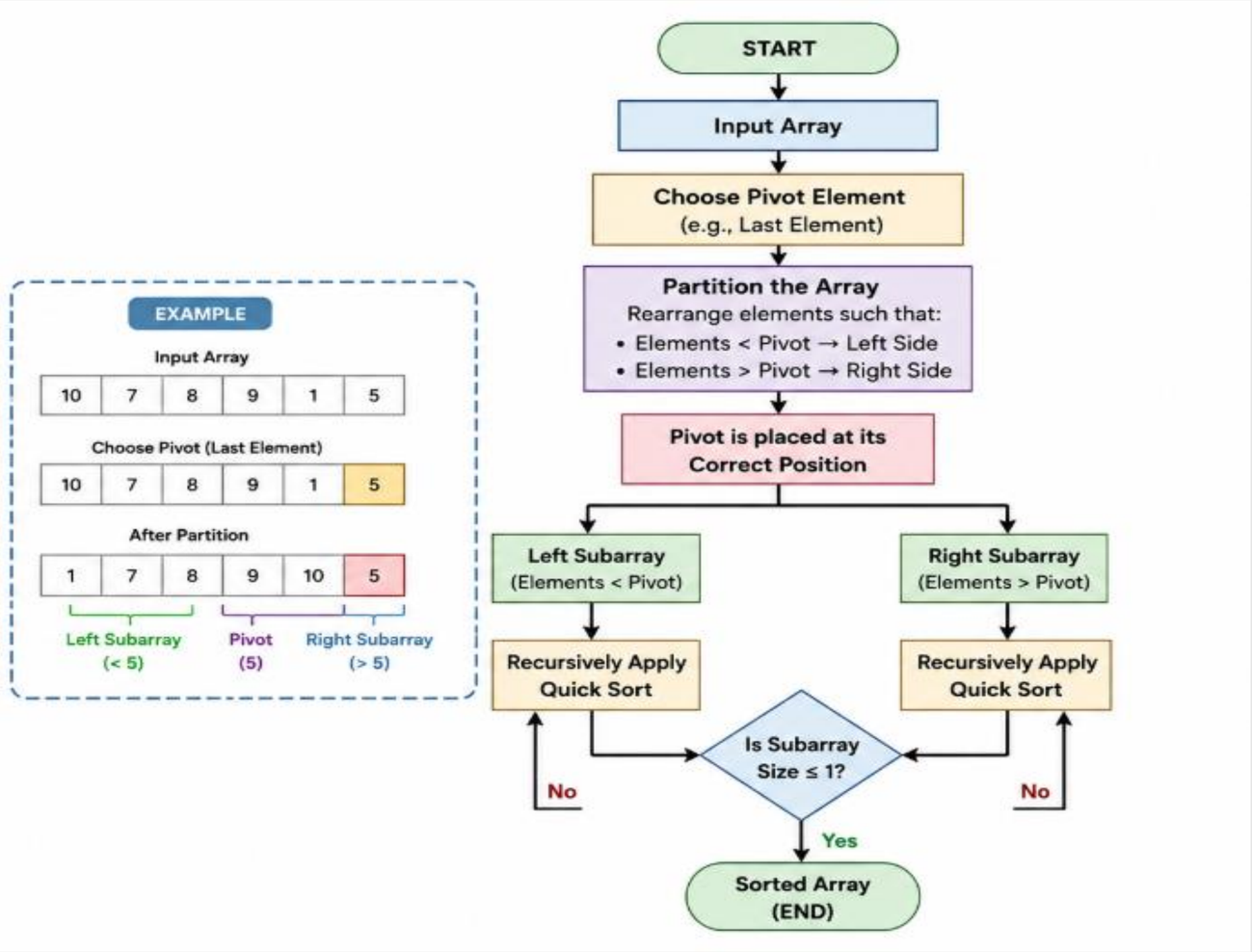
PROJECT STATEMENT

- Analyze the efficiency of the Quick Sort algorithm under Best, Average, and Worst Case scenarios.
- Identify the input conditions that lead to each case.
- Study the effect of different pivot selection strategies on performance.
- Compare the time complexity using asymptotic notation.
- Determine which case has the greatest impact on real-world applications

ALGORITHM

- Algorithm: Quick Sort
- Technique: Divide and Conquer
- Pivot Selection: Last Element
- Sorting Type: In-place Comparison Sort
- Working: Partition → Recursion → Sorted Array
- Average Time Complexity: O(n log n)

SYSTEM ARCHITECTURE



SAMPLE CODE & OUTPUT

```c
#include <stdio.h>
void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[], int low, int high)
{
    int pivot = arr[high];
    int i = low - 1;
    for (int j = low; j < high; j++)
    {
        if (arr[j] < pivot)
        {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return i + 1;
}
void quickSort(int arr[], int low, int high)
{
    if (low < high)
```

```
Enter the number of elements: 6
Enter 6 elements:
71
25
68
413
7118
12

Sorted Array:
12 25 68 71 413 7118

=== Code Execution Successful ===
```

COMPLEXITY ANALYSIS

Case	Input Condition	Pivot Selection	Time Complexity
Best Case	Pivot divides array equally	Middle/Median Pivot	O(n log n)
Average Case	Random input	Random Pivot	O(n log n)
Worst Case	Already sorted or reverse sorted	First/Last Pivot	$O(n^2)$

CONCLUSION

- Its average-case time complexity is O(n log n), making it suitable for most applications.
- The best case occurs when the pivot divides the array into balanced partitions.
- The worst case $O(n^2)$ occurs due to poor pivot selection, such as already sorted data.
- Choosing a random or median pivot helps improve performance and reduces the chance of the worst case.
- Quick Sort is widely used because of its fast average performance and low memory usage.

REFERENCES

- THOMAS H. CORMEN, CHARLES E. LEISERSON, RONALD L. RIVEST, AND CLIFFORD STEIN, INTRODUCTION TO ALGORITHMS, 4TH EDITION, MIT PRESS.
- Ellis Horowitz, Sartaj Sahni, and Susan Anderson-Freed, Fundamentals of Data Structures in C.
- Mark Allen Weiss, Data Structures and Algorithm Analysis in C.
- GeeksforGeeks – Quick Sort Tutorial.
- Programiz – Quick Sort Algorithm.
- TutorialsPoint – Data Structures and Algorithms.

THANK YOU

CO1-AT3

TECHNICAL PRESENTATION

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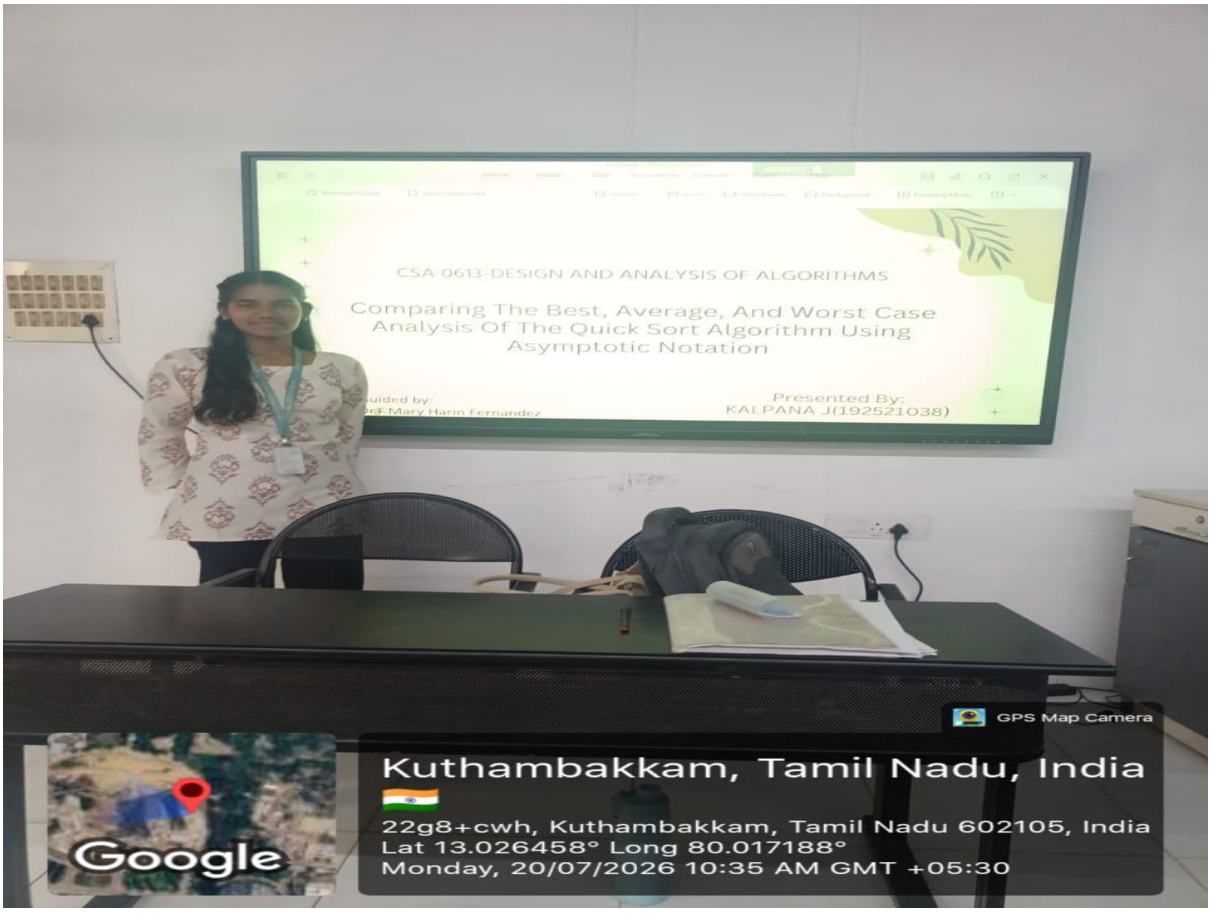

Quick Sort is a divide-and-conquer sorting algorithm whose performance varies depending on how the pivot divides the array. In the best case, the pivot splits the array into two equal halves, resulting in a time complexity of O(n log n). In the average case, the partitions are reasonably balanced, so the algorithm also runs in O(n log n). In the worst case, the pivot always becomes the smallest or largest element, creating highly unbalanced partitions, and the time complexity increases to $O(n^2)$. Using asymptotic notation (Big-O, Theta, and Omega), these cases help analyze the algorithm's efficiency for different input scenarios, showing that Quick Sort is generally one of the fastest sorting algorithms in practice, despite its worst-case behavior.